An Examination of Moderator Effects in the Four-Stage Loyalty Model

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Oliver's 1997 four-stage loyalty model proposes that loyalty consists of belief, affect, intention, and action. Although this loyalty model has recently been subject to empirical examination, the issue of moderator variables has been largely neglected. This article fills that void by analyzing the moderating effects of selected personal and situational characteristics, using a sample of 888 customers of a large do-it-yourself retailer. The results of multigroup causal analysis suggest that these moderators exert an influence on the development of the different stages of the loyalty sequence. Specifically, age, income, education and expertise, price orientation, critical incident recovery, and loyalty card membership are found to be important moderators of the links in the four-stage loyalty model. Limitations of the study are outlined, and implications for both research and managerial practice are discussed.

Keywords: *customer loyalty; four-stage loyalty model; moderator effects; structural equation modeling*

Numerous studies have linked customer satisfaction to financial outcomes (e.g., Anderson, Fornell, and Lehmann 1994; Anderson, Fornell, and Rust 1997; Bernhardt, Donthu, and Kennett 2000; Ittner and Larcker 1998; Keiningham et al. 1999). Yet contemporary marketing thought, turning from a transaction orientation to a relationship orientation (Berry 1995; Grönroos 1995; Morgan and Hunt 1994), acknowledges that gaining and sustaining customer loyalty as the ultimate goal may be more important than achieving customer satisfaction (Agustin and Singh 2005). The relationship between satisfaction and profit, for instance, was examined in two early studies by Fornell and Wernerfeld (1987). They noted a significant impact of complaint handling programs on customer retention and, therefore, on profitability. Obviously, the link between customer satisfaction, customer loyalty, and financial outcome is not as straightforward as it may seem (Carroll 1991; Carroll and Rose 1993; Reinartz and Kumar 2000). Yet researchers and managers acknowledge that small changes in loyalty and retention can yield disproportionately large changes in profitability (Reichheld 1993; Reichheld, Markey, and Hopton 2000; Reicheld and Teal 1996). Therefore, loyalty seems to be an important antecedent to financial outcome.

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Despite this obvious managerial relevance, earlier research has primarily analyzed the link between satisfaction ratings and repurchase intention; few studies have examined the link between satisfaction ratings and repurchase behavior (Bloemer and de Ruyter 1998; Mittal and Kamakura 2001; Zeithaml 2000). Adding to that stream of research, Seiders et al. (2005) summarized and extended the literature by proposing that the relationship between satisfaction and repurchase behavior is moderated by customer, relational, and marketplace characteristics. Only recently has Oliver's (1997) four-stage loyalty model been subject to more extensive empirical testing (e.g., Harris and Goode 2004; Olsen 2002). Our study adds to that stream of research by discussing the links between the four loyalty stages and potential moderators of these links.

In the following, we briefly discuss earlier conceptualizations of the loyalty construct. Oliver's (1997, 1999) framework of the four stages of loyalty is then introduced. After discussing the four stages, the focus of analysis centers on moderator variables affecting the links between the different loyalty stages. Theoretical reasoning and empirical evidence suggest potential moderators for these links. A set of hypotheses is derived and subsequently tested, using structural equation modeling and multigroup analysis. The sample on which we test our hypotheses consists of 888 customers of a large German do-it-yourself (DIY) retailer. In the last section of the article, we discuss implications of our findings, with particular respect to generating a deeper understanding of the conceptual four-stage loyalty model.

CONCEPTUAL FRAMEWORK

Consumer choice behavior is not a random process, and therefore, market shares of exchangeable goods or services are not equally distributed. Preferences in purchase behavior lead to the questions of why and how such preferences emerge. With respect to that observation, the construct of "loyalty" was introduced. Until the 1970s, loyalty was understood as repeat purchase behavior, primarily considering repeat purchase cycles and suggesting a stochastic nature of such cycles (Bass 1974), while largely neglecting phenomena like multibrand loyalty. That shortcoming led to the emergence of a behavioral approach toward explaining purchase patterns and later to the psychological perspective of loyalty.

Jacoby and colleagues (Jacoby and Chestnut 1978; Jacoby and Kyner 1973) were among the first proponents of such a behavioral approach. Loyalty was defined as a biased (nonrandom) repeat purchase of a specific brand (from a set of alternatives) over time by a consumer, using a deliberate evaluation process (Jacoby and Kyner 1973, p. 2). Later, Jacoby and Chestnut (1978) noted that the belief, affect, and intention structure of a consumer must be examined in order to analyze loyalty. On the basis of expectancy-value theory, Jacoby and Chestnut propose that all three evaluative phases must be in place for true loyalty to exist. The authors distinguish between true focal brand loyalty, true multibrand loyalty, nonloyal repeat purchase behavior, and happenstance purchasing. The initial model of loyalty implies that (a) the brand information held by the consumer must clearly identify the focal brand as being superior to competitive offerings (cognition), (b) the consumer must exhibit a greater liking of the focal brand (affect), and (c) the consumer must intend to buy the focal brand as opposed to competitors' offerings (intention) (Jacoby and Chestnut 1978).

Despite these seminal works, there is still no universal agreement on the definition of loyalty (e.g., Dick and Basu 1994; Jacoby and Chestnut 1978; Oliver 1999; Uncles, Dowling, and Hammond 2003). According to Uncles, Dowling, and Hammond (2003), three popular conceptualizations of loyalty exist: loyalty as an attitude that leads to a relationship with the brand; loyalty mainly expressed in terms of revealed behavior (i.e., the pattern of past purchases); and buying moderated by the individual's characteristics, circumstances, and/or the purchase situation. In our study, we use Oliver's (1997) definition of loyalty because it includes both attitudinal and behavioral aspects of the construct. However, we are not trying to imply that this is the best or most elaborate definition of loyalty; rather, for our purpose—identifying and testing for potential moderators between different loyalty stages—it is the most suitable. Oliver's model (1997) explicitly discusses conditions under which loyalty emerges and which factors may hinder such an emergence.

Oliver (1997, p. 392) defined loyalty as a deeply held commitment to rebuy or repatronize a preferred product or service consistently in the future, thereby causing repetitive same-brand or same brand-set purchasing, despite situational influences and marketing efforts that have the potential to cause switching behavior. He introduced a four-stage loyalty model, implying that different aspects of loyalty do not emerge simultaneously but, rather, consecutively over time (Oliver 1999). More than a clarification, this model extends the loyalty sequence "cognitive-affective-conative" by including an observable behavior, for example, actual purchase behavior. Oliver (1999) argued that consumers can be loyal at each phase relating to different elements of the attitude development structure. At each loyalty stage, different factors influencing loyalty can be detected.

Cognitive loyalty. The first loyalty phase is cognitive loyalty. Consumer loyalty at this stage is determined by information of the offering, such as price, quality, and so

forth. It is the weakest type of loyalty because it is directed at costs and benefits of an offering and not at the brand itself. Therefore, consumers are likely to switch once they perceive alternative offerings as being superior with respect to the cost-benefit ratio (Kalyanaram and Little 1994; Sivakumar and Raj 1997). Cognitive loyalty is largely influenced by the consumer's evaluative response to experience, in particular to the perceived performance of an offering relative to price (= value).

Affective loyalty. Affective loyalty relates to a favorable attitude toward a specific brand or product. Attitude itself is a function of cognition (e.g., expectation). Expectancy confirmation leads to satisfaction, which in turn effectuates affective loyalty (Bitner 1990). Oliver (1997) defined satisfaction as "the consumer's fulfillment response, the degree to which the level of fulfillment is pleasant or unpleasant" (p. 28). Satisfaction is a global affect evaluation or feeling state that can be predicted from perceived performance as the cognitive component of the evaluation. Affective loyalty is also subject to deterioration. The deterioration is primarily caused by an increased attractiveness of competitive offerings (Sambandam and Lord 1995) and an enhanced liking for competitive brands, which is, for instance, conveyed through imagery and association used in competitive communications (Oliver 1999).

Conative loyalty. Conative loyalty implies that attitudinal loyalty must be accompanied by a desire to intend an action, for example, repurchase a particular brand. It is stronger than affective loyalty but has vulnerabilities as well. Repeated delivery failures are a particularly strong factor in diminishing conative loyalty. Consumers are more likely to try alternative offerings if they experience frequent service failures. Even though the consumer is conatively loyal, he or she has not developed the resolve to avoid consideration of alternative offerings (Oliver 1999).

Action loyalty. Action control studies imply that not all intentions are transformed into action (e.g., Kuhl and Beckmann 1985). That stream of research concludes that the three previous loyalty states may result in a readiness to act (in this case, to buy). This readiness is accompanied by the consumer's willingness to search for the favorite offering despite considerable effort necessary to do so. Competitive offerings are not considered as alternatives. This facilitator of repurchase is called action inertia (Oliver 1999).

Despite the many attempts to consider selected links between different loyalty phases, relatively little empirical research has been conducted on testing the total four-stage loyalty model (notable exceptions being Harris and Goode 2004, Olsen 2002, and Sivadas and Baker-Prewitt 2000). Our study tries to fill that void by empirically testing Oliver's (1997) loyalty model in a retail setting. At first, we test the three main effects—the links between cognitive-affective, affective-conative, and conative-action loyalty. Then, extending previous research, we test possible moderators affecting the links between the different loyalty phases. These moderating effects are developed in the next section.

MODERATORS IN THE FOUR-STAGE LOYALTY MODEL

In the past decade, a substantial body of research has been conducted to investigate moderating variables influencing the formation of customer loyalty (e.g., Bloemer and Kasper 1995; Homburg and Giering 2001; Homburg, Giering, and Menon 2003).

Based on a review of the literature, potential moderator variables can be roughly divided into two groups, personal characteristics and situational characteristics. The relevance of personal characteristics (e.g., age, gender, education) has not only been found in the context of consumer loyalty (Crask and Reynolds 1978; Korgaonkar, Lund, and Price 1985) but also in different fields of consumer behavior such as decision making (Zeithaml 1985) or purchasing involvement (Slama and Tashlian 1985). The influence of personal characteristics is thus proposed to be more general in nature. Therefore, we expect personal characteristics to be general moderators in the four-stage loyalty model, influencing each link in the model. In contrast, we assume that situational factors (e.g., product expertise, price orientation) solely moderate specific links.

Incorporating previous findings, four personal factors and four situational factors were chosen. A brief discussion of each potential moderator follows.

Personal Characteristics

Age. Wakefield and Baker (1998) argued that age should not only be treated as a predictor variable for satisfaction and loyalty but also as a moderator. Several theories can explain a moderating effect of age on the link between satisfaction and loyalty (Lambert-Pandraud, Laurent, and Lapersonne 2005; Moskovitch 1982). Information processing (Moskovitch 1982), in particular, suggests that older consumers are less likely to seek new information (Wells and Gubar 1966), relying more strongly on heuristic or schema-based forms of processing (Wilkes 1992; Yoon 1997). Therefore, we expect older consumers to rely more on fewer decision criteria, such as their perceived satisfaction with a retailer, whereas younger consumers seek alternative information that might also influence their loyalty. Hence, the relationship between

satisfaction and loyalty is stronger for older consumers than for younger consumers.

Empirical evidence offers some support for these theoretical explanations. Homburg and Giering (2001) found that age moderates the links between satisfaction with the product and loyalty such that these links will be stronger for older consumers. On the basis of theoretical reasoning and empirical evidence, we expect age to affect the strength of the links between the four loyalty stages. Taking this together, the following hypothesis can be formulated:

Hypothesis 1a-c: Age moderates the links between adjacent stages in the loyalty model; specifically, the links will be stronger for older consumers than for younger consumers.

Gender. The impact of gender on response to marketing strategies is generally acknowledged. In addition to possible direct effects of gender on outcome variables such as loyalty (e.g., Fournier 1998; Korgaonkar, Lund, and Price 1985; Odekerken-Schröder et al. 2000), it can be noted that gender is widely used as a moderator variable in marketing in general and in consumer behavior in particular (Saad and Gill 2000). Such a moderating effect can be explained by drawing on social role theory as well as by use of evolutionary psychology. (For a review and comparison of the two approaches, see Archer 1996). According to the above theories, men are more willing to take risks than are women (Powell and Ansic 1997) because, socially, men are expected to behave in such a manner (social role theory) and because this behavior has given men an advantage in the process of natural selection (evolutionary psychology). Because it is riskier to switch a provider and to try something new, men should be less likely to remain loyal if their satisfaction levels change. Conversely, we expect women to react less strongly to satisfaction changes than men do.

Empirical evidence in the context of loyalty shows that gender has been found to moderate the relationship between various aspects of satisfaction and selected measures of loyalty. For instance, Mittal and Kamakura (2001) found that the relationship between satisfaction and repurchase behavior is stronger for men than for women. Similarly, Homburg and Giering (2001) offered some support for such a moderating effect. They were able to show that men who were satisfied with a product are more likely to repurchase it than were women. Based on these findings and the theoretical reasoning outlined above, a moderating effect of gender on the four-stage loyalty model can be assumed:

Hypothesis 2a-c: Gender moderates the links between adjacent stages in the loyalty model; specifically, the links will be stronger for male consumers than for female consumers.

Education. The role of education in the context of loyalty has been given little attention. It is generally acknowledged that people with higher levels of education usually engage more in information gathering and processing and use more information prior to decision making, whereas less well educated people rely more on fewer information cues (Capon and Burke 1980; Claxton, Frey, and Portis 1974). In contrast to people with lower educational attainments, it is argued that better educated consumers feel more comfortable when dealing with, and relying on, new information (Homburg and Giering 2001). We expect, therefore, that better educated consumers seek alternative information about a particular retailer, apart from their satisfaction level, whereas less well educated consumers see satisfaction as an important key information cue on which to base their purchase decision.

Empirical support for a moderating role of education is scarce. Notably, Mittal and Kamakura (2001) found that education moderates the link between satisfaction and retention such that the link is weaker for individuals with more education. On the basis of this finding and on the theoretical reasoning outlined above, we expect that education moderates the links in the four-stage loyalty model:

Hypothesis 3a-c: Education moderates the links between adjacent stages in the loyalty model; specifically, the links will be stronger for less-educated consumers than for well-educated consumers.

Income. A considerable body of research suggests that income is related to consumer loyalty (Crask and Reynolds 1978; Korgaonkar, Lund, and Price 1985; Zeithaml 1985). We expect the link between satisfaction and loyalty to be stronger for low-income consumers than for the wealthier consumers. The reason for that is very much in line with the theoretical explanation offered in the justification of the moderating effect of education. It is generally assumed that people with higher income have achieved higher levels of education (Farley 1964).

Hence, consumers with lower income rely more on their satisfaction with a particular retailer as a key information cue. Therefore, a change in the satisfaction level of a low-income consumer will result in a much stronger change in his or her level of loyalty than for a higher-income consumer. The wealthier consumer will also search for information other than his current satisfaction level to base his repurchase (intention) on.

Homburg and Giering (2001) found partial evidence for such a moderating effect of income on the relationship between satisfaction and loyalty. With respect to these empirical results and the theoretical reasoning, we assume a

moderating effect of income on the four-stage loyalty model:

Hypothesis 4a-c: Income moderates the links between adjacent stages in the loyalty model; specifically, the links will be stronger for low-income consumers than for high-income consumers.

Situational Characteristics

Expertise. Previous research has shown that consumer expertise can influence consumer loyalty (e.g., Alba and Hutchinson 1987; Bettman and Sujan 1987; de Bont and Schoormans 1995; East 1992; Maheswaran 1994; Mitchell and Dacin 1996; Till and Busler 2000). Expert consumers have comprehensive knowledge of the product category and of alternative choices. As a result, they are more aware of and susceptible to competitive offerings than are less experienced consumers (Mitchell and Dacin 1996), who rely more on a limited number of information sources. Scott, Osgood, and Peterson (1979) found that experts with great domain knowledge exhibit greater cognitive differentiation than those with less domain-specific knowledge. That is, expert consumers use more attributes and more attribute levels to differentiate between objects/products in a domain. Therefore, expert consumers base their liking of a retailer on a wide range of information obtained from many sources, whereas less experienced consumers rely more on their evaluation of past experiences with that retailer. Hence, a change in the evaluation of past experiences will have a much stronger effect on affective loyalty for less experienced consumers than for expert consumers. For experts, evaluation of past experiences is just one of many information cues influencing their general liking of a provider. Therefore, we assume that expertise negatively moderates the link between cognitive and affective loyalty.

Hypothesis 5: Customer expertise exhibits a moderating effect on the relationship between cognitive and affective loyalty; specifically, the link will be stronger for less experienced consumers than for well-experienced consumers.

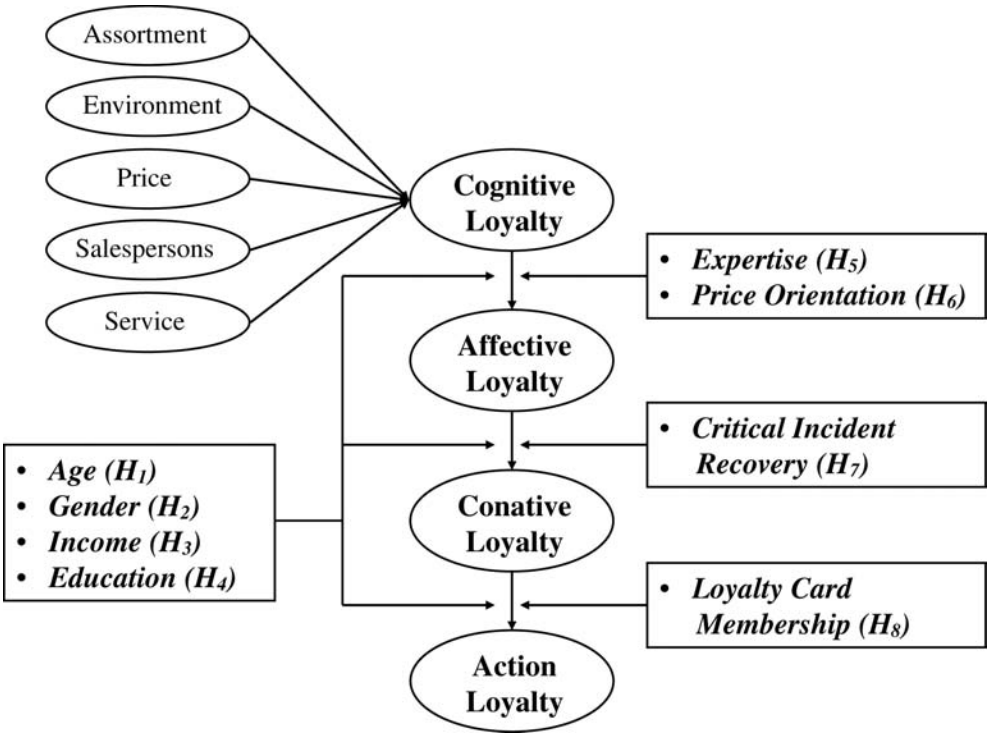
Price orientation. Price orientation is one of the core motivational drivers of shopping behavior. Several authors analyzed the direct and indirect impact of price orientation on outcome variables such as loyalty (Chance and French 1972; Hoyer 1984; Kim, Srinivasan, and Wilcox 1999; Mägi 2003). Price-oriented consumers constantly search for better deals. Therefore, they are more aware of competitive offerings than are less price-oriented consumers. Even if price-oriented consumers have had positive experiences with a specific retailer, they might not automatically become affectively loyal, because searching for better deals is pleasurable to them (Arnold and Reynolds 2003). Hence, for price-oriented consumers, cognitive loyalty is just one factor influencing their affective loyalty. Conversely, less price-oriented consumers do not generally enjoy searching for bargains. They are less aware of competitive offerings and base their affective loyalty much more on their past experiences with the retailer. This reasoning is very much in line with Oliver's (1997) notion that affective loyalty is subject to deterioration caused mainly by an increased perceived attractiveness of competitive offerings. We therefore expect a change in cognitive loyalty toward a provider to have a much stronger impact on affective loyalty for less price-oriented consumers than for price-oriented consumers. Hence, we hypothesize price orientation to negatively moderate the link between cognitive and affective loyalty.

Hypothesis 6: Price orientation exhibits a moderating effect on the relationship between cognitive and affective loyalty; specifically, the link will be stronger for less price-oriented consumers than for more price-oriented consumers.

Critical incident recovery. Critical incidents have received a considerable amount of attention in the services and relationship marketing literature (Bennett 1997; Bolton 1998; Bolton and Lemon 1999; Mattila and Patterson 2004; Odekerken-Schröder et al. 2000). A critical incident can be defined as an encounter that in some respect differs, negatively or positively, from what is expected (Bitner 1990; Flanagan 1954). In that sense, critical incident recovery can be understood as a service provider's effort to make up for a negative critical incident to the full satisfaction of the consumer who experienced that negative critical incident. Consumers who generally like shopping at a certain retailer might differ in their intention to repurchase, depending on the fact that they have or have not experienced a critical incident recovery. In line with Oliver (1997), we draw that conclusion because one of the key conditions for affectively loyal consumers to become conatively loyal is their involvement with the provider. Maxham and Netemeyer (2002, 2003) concluded that consumers who experience a retailer's extra effort to make up for a service failure are highly involved in the purchase process. On the other hand, consumers who have not experienced a service recovery are less involved with the retailer. Therefore, their affective loyalty has a much weaker impact on their repurchase intention than for recovered consumers. On the basis of this reasoning, we assume that in the case of a negative critical incident, a successful service recovery moderates the link between affective and conative loyalty positively.

Hypothesis 7: Critical incident recovery exhibits a moderating effect on the relationship between affective

FIGURE 1
Conceptual Model and Hypotheses



and conative loyalty; specifically, the link will be stronger for recovered consumers than for consumers who have not experienced a critical incident.

Loyalty card membership. Loyalty cards have been implemented by a growing number of retailers; however, empirical proof with regard to their efficiency and effectiveness to increase customer retention remains inconclusive (Bellizzi and Bristol 2004; Mägi 2003; Reinartz and Kumar 2000). Still, loyalty cards build a kind of financial switching barrier (Burnham, Frels, and Mahajan 2003), representing sunk costs for the consumer. There is empirical evidence that higher switching costs positively influence customer loyalty (Jones, Mothersbaugh, and Beatty 2002). Besides possible direct effects on loyalty, we suspect that loyalty card membership moderates the link between conative loyalty and action loyalty. Comparing two consumers who intend to purchase at a certain retailer, with one holding a loyalty card, whereas the other does not, the card holder is much more likely to purchase at that retailer because the consumer does not want to lose the benefit of a loyalty card or of the investment (e.g., time and effort) in getting such a loyalty card. Hence, the link between conative and action loyalty is stronger for loyalty card holders. This reasoning is again in line with Oliver (1997), who noted that key sustainers of action loyalty are, in fact, sunk costs, and actual purchase will be more likely for consumers faced with sunk costs as opposed to those with no sunk costs associated with switching. In this respect, a loyalty card might serve as an aid to transform intention into action. Therefore, we expect loyalty card membership to positively moderate the relationship between conative and action loyalty.

Hypothesis 8: Loyalty card membership exhibits a moderating effect on the relationship between conative and action loyalty; specifically, the link will be stronger for consumers holding a loyalty card than for consumers who do not hold such a loyalty card.

Figure 1 provides an overview of the three hypothesized main effects in the four-stage loyalty model and of the eight moderator effects.

METHOD

Data Collection Procedure and Sample

The German DIY market is highly competitive, and the Top 10 retailers account for roughly 80% of total sales.

The market grows slowly, less than 2% a year, and the average profit margin is about 3% (Metro Group 2005, p. 24). Moreover, there is an ongoing threat of new market entries, especially from large international DIY chains. The retailer in our study is quite representative for this market. Because of guaranteed confidentiality of our data, we cannot disclose further details related to the identity of the retailer.

The sample in our study consisted of 888 shoppers from a large German DIY franchise system, and customers were intercepted and interviewed inside 10 different franchise outlets. The franchise outlets were selected by the franchisor, who assured that they were representative of the system as a whole. In particular, franchise outlets were representative in regard to size, performance, and customer base.

All interviews with customers were processed within a given time period of 12 days in October 2003. The interviewers were instructed to follow quota on age and gender, based on information given by the retailer on their customers and on DIY customers in general. In following the quota, the sample drawn was representative not only of that particular retailer but of the German DIY market in general.

During the interview, no respondent showed any signs of difficulty understanding or answering the questions. A total of 1,000 customer surveys were returned by the interviewers, evenly divided among the 10 franchise outlets (100 each). Because of missing values, 888 from the total of 1,000 questionnaires remained for further analysis. The proportion of unusable questionnaires was approximately equally distributed among the 10 outlets. The final sample consisted of 335 women and 553 men.

Because of legal restrictions in Germany, it was impossible to collect objective data on action loyalty in particular (e.g., loyalty card data). Therefore, there is the possibility of common-method bias (CMB) in our study because we have only one source (one customer) that gives his assessment of the dependent and independent variables. As procedural remedies cannot completely eliminate CMB, we applied the methods suggested by Podsakoff and colleagues (2003) to test for CMB. In line with their reasoning, we analyzed our data using the "single-method-factor approach." We estimated the model with a single-method first-order factor added to the indicators of the four loyalty stages. Although the fit indices slightly improved when adding a common-method factor (Goodness-of-Fit Index [GFI] = .927 [basic model: .902]; Adjusted Goodness-of-Fit Index [AGFI] = .905 [model: .878]; root mean indicator RMR = .035 [model: .043]; root mean square error of approximation [RMSEA] = .052 [model: .063]), results reveal that even with common-method variance controlled, the three path coefficients between the loyalty stages remain significant, and the proportion of variance explained was almost unchanged in both models. Therefore, we conclude that CMB is not a significant issue in our study.

Measures and Measurement Properties

Measurement of the Constructs

Conceptualization and items for measuring the four constructs of the model were developed, drawing on prior research in the quality-satisfaction-loyalty literature. Except for cognitive loyalty, which was operationalized as a five-dimensional construct, the loyalty constructs were measured using multi-item scales adapted from previous studies. All items were measured using a 5-point Likert-type scale anchored by 1 (*very satisfied*, *like very much*, or *totally agree*) and 5 (*very dissatisfied*, *dislike very much*, or *totally disagree*).

Cognitive loyalty. Cognitive loyalty was defined and measured as evaluation of attribute performance associated with the retail outlet. In line with Westbrook's (1981) operationalization of retail satisfaction, the respondents were asked to rate the perceived performance of the five dimensions of assortment, environment, price, salesperson, and service on 5-point Likert-type scales.

Affective loyalty. Affective loyalty was defined as a person's global affect evaluation or feeling state. As proposed by Oliver (1997), we used *overall satisfaction* and *liking* to cover affective loyalty. Both items were measured using a 5-point Likert-type scale.

Conative loyalty. Conative loyalty was defined as a customer's behavioral intention to continue buying at a retail store in the future, accompanied by a deep commitment to that store. The scale used in the current study was adapted from the behavioral intention battery developed by Zeithaml, Berry, and Parasuraman (1996) and included three items: willingness to recommend, cross-buying intention, and repurchase intention. Again, a 5-point Likert-type scale ranging from 1 (*completely agree*) to 5 (*completely disagree*) was used.

Action loyalty. Action loyalty in this article focuses on (purchase) behavior. Hence, we do not focus on attitudes such as an intention to purchase or intention to overcome an obstacle. Harris and Goode's (2004) article is one of the few studies using Oliver's four-stage framework. Just as in our study, the authors operationalize action loyalty as displayed choice preference and not as overcoming obstacle. Also in line with Oliver (1997), action loyalty includes habit and routinized response behavior. The most common

TABLE 1
Descriptive Statistics

Item	M	SD
Cognitive loyalty		
Assortment		
Assortment of departments	1.891	0.548
Quality level	1.936	0.547
Merchandise selection	1.857	0.605
Well-stocked departments	2.126	0.672
Environment		
Interior appearance	2.127	0.638
Store layout	2.003	0.708
Roominess	1.993	0.613
Cleanliness	1.731	0.518
Price		
Special sales	2.426	0.696
Price level	2.301	0.640
Store salespersons		
Helpfulness of salespeople	1.835	0.772
Friendliness	1.585	0.640
Politeness	1.559	0.632
Service		
Fairness of adjustment	1.761	0.561
Willingness to exchange	1.658	0.549
Affective loyalty		
Overall satisfaction	1.966	0.592
Liking	2.023	0.615
Conative loyalty		
Repurchase intention	1.725	0.612
Cross-buying intention	1.674	0.729
Word-of-mouth behavior intention	1.205	0.548
Action loyalty		
Word-of-mouth behavior	1.601	1.117
Purchase frequency	3.075	1.240
Money spent (per year)	2.768	1.403

NOTE: All variables range from 1 to 5, where 1 represents the strongest possible (e.g., *very satisfied* or *very likely*) and 5 the weakest possible (e.g., *very dissatisfied* or *very unlikely*).

assessments of behavioral loyalty are repurchasing patterns or behavioral frequency. For our purposes, we used three self-reported behavior measures: word-of-mouth behavior in the past, purchase frequency, and actual money spent at this particular retailer per year. The behavior measures were selected to reflect the intention measures mentioned above and to capture true and not spurious loyalty purchase patterns. Unfortunately, because of legal restrictions, we were unable to use loyalty card data to reflect actual purchase behavior.

Finally, the eight moderating factors were assessed by single-item measures (see the appendix for a list of the scale items used in this study, Table 1 for descriptive statistics, and Table 2 for correlations between constructs).

Reliability and Validity

Measurement reliability was examined through confirmatory factor analysis and the calculation of Cronbach's alpha coefficient (Cronbach 1951). Analyses were conducted separately for the constructs. The appendix shows the values of composite reliability gained on the basis of confirmatory factor analysis and the values of coefficient alpha. It can be noted that the coefficient alpha is lower than .7, the threshold generally proposed in the literature (Nunnally 1978), for assortment, affective loyalty, conative loyalty, and action loyalty. Nevertheless, because composite reliabilities for all constructs are larger than .6 (Bagozzi and Yi 1988), we did not change any measurement model. It is generally acknowledged that composite reliability is a better and stronger measure for scale reliability than coefficient alpha (e.g., Bagozzi and Yi 1988; Homburg and Giering, 2001; Homburg, Giering, and Menon 2003). There are two main reasons. At first, the alpha value depends highly on the number of scale items. With only a few items for the scales, it is not surprising that the values are rather low. Moreover, composite reliability accounts for measurement errors, whereas coefficient alpha assumes error-free measures. Taking this together, we feel confident that our scales are reliable even if alpha values do not meet the generally proposed thresholds.

As proposed, cognitive loyalty consists of the five dimensions of assortment, environment, price, salesperson, and service, whereas affective, conative, and action loyalty are unidimensional. Validity of the cognitive loyalty construct was assessed using the criterion proposed by Fornell and Larcker (1981, p. 46). Again, the criterion was met. Finally, to confirm that the four loyalty constructs are distinct from each other, we tested whether the interconstruct correlation was significantly different from 1 (Bagozzi, Yi, and Phillipps 1991) through the analysis of chi-square difference tests. The chi-square difference test is conducted by estimating the fit of a model when the interconstruct correlation is constrained to 1 and then estimating the same model with the interconstruct correlation free to be estimated. The test statistic is the difference in chi-square values with one more degree of freedom. In the present study, all changes in chi-square were significant at a .01 level, indicating that improved fits are obtained by separating the constructs. Therefore, reliability and validity of the constructs in this study are acceptable.

Analysis

At the first stage of analysis, the main effects in the four-stage loyalty model were tested using structural equation modeling.[1] Next, the influences of moderator variables on the main effects were analyzed. To investigate moderating effects, we conducted median splits based on

TABLE 2
Correlation Matrix

	Assortment	*Environment*	*Price*	*Store Salespersons*	*Service*
Assortment	1.000				
Environment	.719	1.000			
Price	.433	.477	1.000		
Store salespersons	.353	.340	.237	1.000	
Service	.352	.302	.312	.401	1.000
Average variance extracted	.549	.601	.640	.692	.810
	Cognitive Loyalty	*Affective Loyalty*	*Conative Loyalty*	*Action Loyalty*	
Cognitive loyalty	1.000				
Affective loyalty	.651	1.000			
Conative loyalty	.351	.459	1.000		
Action loyalty	.069	.092	.204	1.000	
Average variance extracted	.514	.486	.355	.428	

TABLE 3
Path Coefficients

Basic Model Effect	*Path Coefficient*
Cognitive Loyalty → Affective Loyalty	.749
Affective Loyalty → Conative Loyalty	.464
Conative Loyalty → Action Loyalty	.201

NOTE: All path coefficients are significant at .01 level.

the value of the moderator variables (Baron and Kenny 1986). Thus, each moderator variable was divided into high and low groups. In the case of dichotomous variables (critical incident recovery, loyalty card, and gender), we constructed two groups of equal size by random selection.

Using multigroup causal analysis, the test for moderating effects was conducted. First, we computed a nonrestricted model and then restricted the path under investigation to be equal across subgroups. A moderating effect exists if the change in the chi-square value is significant. With one more degree of freedom for the restricted model, the critical value of chi-square differences is 3.84 ($p = .05$).

RESULTS

Main Effects

To test the main effects in the four-stage loyalty model, the goodness-of-fit statistics of the basic model were assessed. GFI is .902, AGFI is .878, RMR is .043, and the probabilistic RMSEA is .063. Although AGFI is slightly below the threshold of .9 generally proposed (Baumgartner and Homburg 1996; Hair et al. 1998), all other indicators point to an acceptable fit. Thus, it is concluded that the model fits the data reasonably well.

Table 3 shows the parameter estimates of the model on the full data set. Path coefficients between cognitive-affective loyalty, affective-conative loyalty, and conative-action loyalty are positive in each case and statistically significant at the .01 level. Hence, all the hypothesized relationships in the four-stage loyalty model were supported by the findings. As in earlier studies of the four-stage loyalty model, the strongest relationship was observed between cognitive and affective loyalty (Harris and Goode 2004; Sivadas and Baker-Prewitt 2000). Also in line with previous findings, the link between conative and action loyalty showed the lowest intensity (.201, $p < .01$).

Moderating Effects

After confirming the influence of the three postulated relationships, we tested for moderator effects. In a first step, an overall chi-square difference test was conducted for the personal characteristic variables. Here, a model that imposes equality constraints on all three paths across subgroups was compared with the general nonrestricted model. Hence, we tested the null hypothesis that the respective moderator variables have no influence on the relationships in the four-stage loyalty model. With three more degrees of freedom, the restricted model exhibits a significant chi-square difference (at $p < .05$) for three of four effects: age, education, and income (Table 4), but not for gender. Whereas previous studies in the satisfaction-loyalty research support the moderating role of gender (e.g., Homburg and Giering 2001), in the current study, gender showed no moderating effect (chi-square difference with

TABLE 4
Results of Multigroup Analysis

Personal Characteristic	*Age*		χ^2	$\Delta\chi^2$(df = *1*)
	Young	*Old*		
Cognitive Loyalty → Affective Loyalty	.712	.758	1680.2	9.98**
Affective Loyalty → Conative Loyalty	.477	.447	1670.2	0
Conative Loyalty → Action Loyalty	.188	.181	1670.2	0
	$\Delta\chi^2$(*df* = 3): 9.98**			
	Income			
	Low	*High*	χ^2	$\Delta\chi^2$(df = *1*)
Cognitive Loyalty → Affective Loyalty	.896	.691	1,529.0	8.40**
Affective Loyalty → Conative Loyalty	.525	.418	1,520.6	0.02
Conative Loyalty → Action Loyalty	.243	.187	1,521.8	1.22
	$\Delta\chi^2$(df = 3): 9.97**			
	Education			
	Low	*High*	χ^2	$\Delta\chi^2$(df = *1*)
Cognitive Loyalty → Affective Loyalty	.883	.628	1,653.5	16.7**
Affective Loyalty → Conative Loyalty	.521	.385	1,636.9	0.17
Conative Loyalty → Action Loyalty	.200	.184	1,637.2	0.47
	$\Delta\chi^2$(df = 3): 18.36**			

** Significant at .01 level.

three degrees of freedom is 4.96). The remaining three personal characteristics moderate the relationships in our loyalty model such that the links between the loyalty stages are stronger for older, less educated customers with lower incomes. Results are in line with findings in the literature (e.g., Chance and French 1972; Homburg and Giering 2001; Lambert-Pandraud, Laurent, and Lapersonne 2005; Lumpkin and Greenberg 1982; Mittal and Kamakura 2001; Zeithaml 1985). Our results support the finding that the personal characteristics of age, education, and income, in general, are relevant moderators in the context of the four-stage loyalty model.

Next, we analyzed the specific moderating effects of the three personal variables identified above by examining each path of the model separately.

With regard to the findings presented in Table 4, it can be noted that the three personal characteristics of age, education, and income show only one out of three effects—they moderate the link between cognitive and affective loyalty. Hence, only Hypotheses 1a, 3a, and 4a are supported by the data at hand.

Table 5 shows the results of the specific moderating effects of the four situational characteristics: expertise, price orientation, critical incident, and loyalty card membership, as proposed in Hypotheses 5 through 8.

At first, it can be stated that expertise exhibits a moderating effect on the relationship between cognitive and affective loyalty, supporting Hypothesis 5. Expertise moderates the strength of the relationship in such a way that, for customers with low expertise, the link is significantly stronger than for professional customers.

Furthermore, the results indicate that price orientation is a moderator of the link between cognitive and affective loyalty. Hence, Hypothesis 6 is supported. In the case of highly price-oriented customers, the link between cognitive and affective loyalty is less strong than for customers with low price orientation.

As proposed in Hypothesis 7, critical incident recovery moderates the influence from affective to conative loyalty. A chi-square difference of 18.88 indicates strong support for Hypothesis 7. It is worth mentioning that we only looked at those critical incidents that were solved to the satisfaction of the customer, meaning that we were looking at recovered customers. With that in mind, recovered customers who are affectively loyal express stronger behavioral intentions, for example, speaking to friends about their positive experience with that retailer, than do customers who did not experience a critical incident. Unfortunately, we could not compare those customers who experienced a negative critical incident with those having experienced a positive critical incident, because the former group was too small for multigroup analysis, and traditional moderated regression analysis led to insignificant results. That observation is in line with findings on customer-recovery processes (e.g., Halstead and Page 1992; Spreng, Harrel, and Mackoy 1995).

TABLE 5
Results of Multigroup Analysis

	Expertise			
	Low	*High*	χ^2	$\Delta\chi^2$(df = *1*)
Cognitive Loyalty → Affective Loyalty	.795	.713	1,611.899	5.09**
	Price Orientation			
	Low	*High*	χ^2	$\Delta\chi^2$(df = *1*)
Cognitive Loyalty → Affective Loyalty	.809	.747	1,718.97	4.70*
	Critical Incidence Recovery			
	Yes	*No*	χ^2	$\Delta\chi^2$(df = *1*)
Affective Loyalty → Conative Loyalty	.533	.481	1,306.61	18.88**
	Loyalty Card Membership			
	Yes	*No*	χ^2	$\Delta\chi^2$(df = *1*)
Conative Loyalty → Action Loyalty	.322	.270	1,115.14	7.26*

* Significant at .05 level. ** Significant at .01 level.

The last moderator we considered was loyalty card membership, which is expected to moderate the influence of conative to action loyalty. Results indicate support for Hypothesis 8. According to these results, loyalty card owners who intend to buy at a particular retailer are more likely to actually purchase there than are those customers who do not own a loyalty card.

DISCUSSION

Gaining customer loyalty is less straightforward and more complex than several previous studies have suggested. Using Oliver's (1997) model of loyalty, it is not until a customer shows a high consistency through the four distinct stages that loyalty is achieved. In that case, customers possess not only a stable disposition but also a stable, routinized purchase pattern, making them resistant to competitive offers. Therefore, examining moderator variables of the links in the four-stage loyalty model leads to a clearer understanding of how to build and maintain customer loyalty.

First, it can be noted that gender does not moderate any of the links in the four-stage loyalty model. Obviously, and contrary to findings from other researchers (e.g., Homburg and Giering 2001), the links between the four stages are equally strong for both men and women. A reason for that could be that DIY stores offer mostly low-involvement goods. In general, DIY products are purchased less impulsively and with lower levels of affection. Consumers—male or female—might allocate little time and effort to search for DIY products like nails, screws, varnish, or paint. Therefore, insignificant differences in the strength between the four loyalty stages are plausible. Particularly for high-involvement products, we would expect a moderating role of gender, as a study in the context of automobiles has already revealed (Mittal and Kamakura 2001). In this case, the link between satisfaction and loyalty in the product category "automobiles"—a high-involvement category for most men—is stronger for men than for women. However, we accept an alternative view that some DIY products might be high-involvement products as well. Because we have no further details about the products purchased by the respondents of our study, we acknowledge the need for further research to analyze the moderating roles of gender and involvement, and the interaction between these two variables.

The three personal characteristics of age, education, and income exhibit a moderating effect only on the link between cognitive and affective loyalty. For older customers, for customers with lower income, and for customers with lower levels of education, the link from cognitive to affective loyalty is significantly higher than for young, high-income, and well-educated customers. This indicates that the affect evaluation of older, less wealthy, and less educated customers is obviously affected more by the perceived attribute performance of the retailer. The perceived cost-benefit ratio is essential for the satisfaction rating of older, less wealthy, and less educated customers and, hence, for their affective loyalty. Moreover, our results of-

fer some additional insights. Personal characteristics do not seem to have a moderating effect for later stages in the loyalty chain, in particular between conative and action loyalty. Therefore, we suggest that management needs to acknowledge that demographic variables are not sufficient to cluster customers if they want to turn satisfied customers into action-loyal customers.

As proposed in Hypotheses 5 through 8, there is a moderating effect of expertise, price orientation, critical incident recovery, and loyalty card membership. For less experienced customers and customers with a lower price orientation, the relationship between cognitive and affective loyalty is stronger than for expert customers and for customers with a higher price orientation. Less experienced customers with a lower price orientation might rarely shop for DIY products. They have limited price knowledge and might be surprised by the services offered in the store. For these customers, the impact of perceived performance on their feeling state is stronger, leading to higher affective loyalty (Table 5). Thus, our results indicate that customer segments can be identified that are more or less affectively loyal, given the same perceived performance rating. Based on these findings, an important managerial implication is that for price-oriented and expert customers, attempts at increasing affective loyalty through attribute performance improvements are less successful than for novices and customers with low price orientation. Regarding experts and price-oriented customers, further research is needed to evaluate those antecedents that enhance liking and that build up affective loyalty.

The results of our study provide support for the view that critical incident recovery is a strong positive moderator of the relationship between affective and conative loyalty. A customer who generally likes the retailer and then has experienced a successfully resolved critical incident is much more committed to buying at that retailer and is less vulnerable to counterargument by the competitors. Yet, contrary to previous studies suggesting that firms may not always benefit from service recovery efforts over and above a moderate level (e.g., Maxham 2001), our findings point out that service recovery is indeed important to develop customer loyalty. Actions that are taken by retailers in response to service failure positively influence the link between affective loyalty and customer repurchase intentions, cross-buying intentions, and positive word-of-mouth intentions. Thus, our results emphasize the importance of developing service recovery strategies as part of customer loyalty programs.

A loyalty card is intended to be a loyalty builder. Our results indicate that customers who are conatively loyal to a certain retailer are much more likely to actually buy at and recommend that retailer if they hold a loyalty card. That positive effect of loyalty cards must be seen with respect to the growing investments of the retailer in loyalty programs and the mixed results of previous studies concerning the desired outcomes of such programs (for an overview, see, e.g., Mägi 2003; Verhoef 2003). The management must evaluate the potential costs against potential benefits to decide whether or not loyalty cards have positive effects for the company (Verhoef 2003; Verhoef, Franses, and Hoekstra 2002). For the retailer of this study, the results point to a positive moderating effect of loyalty cards, aiding in transforming a customer's repurchase intention into actual repurchase.[2] This insight is important when designing and evaluating the outcomes of loyalty card programs. Thus, although loyalty cards do not affect customer loyalty by enhancing perceived performance or satisfaction, they do strengthen the relationship between an individual's buying intention and the actual buying behavior.

LIMITATIONS AND FURTHER RESEARCH

As with all empirical studies, our study suffers from limitations. First, our object of analysis was a particular retailer from one industry, the German DIY market. The advantage of choosing one service area was that it allowed us to detect context-specific moderating effects. Second, we analyzed data from one particular point in time only. By doing that, we presume there is no time lag between a customer's feeling of being satisfied and the behavioral and attitudinal consequences. Third, we only considered linear relations between the four loyalty stages. However, as indicated in the literature (e.g., Agustin and Singh 2005; Anderson and Mittal 2000; Keiningham, Perkins-Munn, and Evans 2003), these relationships might be curvilinear. Further research is needed to control for these effects in future studies that are conducted in different environmental contexts. As mentioned above, it can be noted that in our study such curvilinear effects cannot better explain the relationships between the different loyalty stages than linear relationships.

Finally, we were unable to analyze actual purchasing behavior because of legal restrictions in using customer-card data. Therefore, we had to use as a proxy the customers' evaluations of two statements on their repurchase behavior.

However, we do not expect severe limitations in the generalization or the substantive implications of our results. The retailer we have analyzed is typical in size and success for the German, or in fact the European, DIY market. Still, we would call for replications of this study in different environmental and cultural contexts. Moreover, longitudinal data would further improve our understanding of the mechanisms influencing the link between satis-

faction and loyalty in general and the effect of certain moderators in particular.

In our study, we did not find any moderating effect of gender. Because this is not in line with previous research, an area for future research could be to investigate if the links between adjacent stages in the loyalty model might not be equally strong for men and women for certain product categories (home appliances versus gardening articles such as flowers). One might speculate that for each gender group, the links between the loyalty stages are themselves moderated by the product category.

In all, the specific moderating effects indicate that research on the loyalty construct is not at all complete and that Oliver's (1997) model gives many hints to further investigate.

APPENDIX

Scale Item[a]	*Alpha*	*Composite Reliability*
Cognitive Loyalty (adapted from Westbrook 1981)		
Performance ratings on		
Assortment	.68	.68
Assortment of departments		
Quality level		
Merchandise selection		
Well-stocked departments		
Environment	.72	.73
Interior appearance		
Store layout		
Roominess		
Cleanliness		
Price	.78	.78
Special sales		
Price level		
Store salespersons	.85	.87
Helpfulness of salespeople		
Friendliness		
Politeness		
Service	.94	.90
Willingness to exchange		
Fairness of adjustments		
Affective loyalty (adapted from Oliver 1997)	.67	.68
Overall satisfaction		
Liking		
Conative loyalty (adapted from Zeithaml, Berry, and Parasuraman 1996)	.69	.61
Repurchase intention		
Cross-buying intention		
Word-of-mouth behavior intention		
Action loyalty	.63	.64
Word-of-mouth behavior		
Purchase frequency		
Money spent (per year)		
Moderator variables		
Gender		
Age		
Education (no educational attainment, high school, college, university)		
Income (total household income in Euros/month)		
Expertise (no to minor expertise, moderate, good, very good)		
Price consciousness		
Critical incident recovery (yes/no)		
Loyalty card membership (yes/no)		

a. All loyalty items were measured using 5-point Likert-type scales, where 1 represents the strongest possible (e.g., *very satisfied* or *very likely*) and 5 represents the weakest possible (e.g., *very dissatisfied* or *very unlikely*). The indicator "money spent per year" was the absolute value transformed into a 5-point scale.

NOTES

1. We also tested for possible curvilinear effects between the four loyalty stages using regression analysis. The model fit and the variance explained did not increase significantly when using nonlinear regressions with single indicators. We thank one anonymous reviewer for raising that point.

2. Note that we do not know if a customer of this particular retailer is also a member of different loyalty card programs from another retailer. We thank one anonymous reviewer for pointing out this limitation.

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